BUILDATHON 2026 · INDIVIDUAL ENTRY

ResearchOS

From research idea to validated result, for the whole team, with an AI co-pilot at every step.

Built by Saurrav Dasgupta · research-os-beige-five.vercel.app · github.com/worldofnisho-471/ResearchOS

WHO IT'S FOR

Months of work, and no one can say if it's on track.

Picture a **[second-year PhD student in environmental engineering]**. Every week she runs experiments and updates a spreadsheet only she reads. Last [month], [she found her three-week result had already been published]. Today she pieces it together from Google Scholar, Excel and WhatsApp updates to her supervisor, which costs her [hours each week and a lot of stress].

THEN

She opens ResearchOS, checks her idea, gets a plan, logs a result, and in [minutes] knows exactly what to change next.

THE PROBLEM

Nothing connects the idea, the plan and the results.

How they solve it today	What it does well	The gap
Google Scholar, AI search tools	Finds papers	No plan, no tracking, no verdict on the idea
Spreadsheets, Notion, notebooks	Records experiments	No judgement; results drift from the hypothesis
Supervisor meetings	Expert judgement	Weekly at best; context gets lost in between

WHY NOT JUST USE CHATGPT?

ChatGPT answers a question. ResearchOS remembers the plan, the results and the evidence, and turns them into your next decision.

Real papers only

Pulled from Semantic Scholar or OpenAlex, and linked. The AI can't cite anything else.

Shared memory

The plan, results and notes persist for the whole team, not one chat.

Evidence over opinion

Scores show their assumptions and which ones are proven.

THE SPEC

One research team. One journey. Five screens.

1
Idea check
Verdict, real papers, assumptions

→

2
Research plan
Tasks with measurable goals

→

3
Log results
Did it meet the goal?

→

4
Dashboard
On track, at risk or off track

→

5
Next steps
AI edits the plan

AHA MOMENT

A result misses its target, and the AI rewrites the plan with the reason why.

OUT OF SCOPE

Grant writing, Gantt charts, instrument data, statistics, marketplaces

HOW THE AI WORKS

Grounded in real data, approved by a human.

1 Search

Semantic Scholar returns the 8 most relevant papers; OpenAlex steps in if it's busy.

2 Assess

The model scores the idea using only those papers, and lists the assumptions behind its score.

3 Track

Every task has a measurable success criterion; each result is logged against it.

4 Review

The AI reads results and lab notes, proposes edits, and a Lead approves them.

Idea scores below 50? Related directions are suggested automatically, built on the team's ORCID publications and ranked with OpenAlex paper trends, NIH funding and web market signals, each with its source.

Every AI step runs as a background job, so a slow model never loses your work. API keys stay on the server; *planned:* a paid AI tier whose providers don't train on your data.

LIVE DEMO · 4 MINUTES

One idea, two people, four minutes.

1. The Lead checks an idea: it scores below 50, with real papers
2. Related directions appear; the Lead starts one and generates its plan
3. The Lead assigns a task, and the Researcher's bell lights up
4. The Researcher moves the card and logs a missed result; both boards update live
5. The Researcher @mentions the Lead; Next steps proposes changes; the Lead approves

Try it yourself: **research-os-beige-five.vercel.app** · demo login **user1@demo.com / demotest1**

JIRA AND CONFLUENCE FOR RESEARCH

Built for the whole research team, not one person’s chat.

Work together live

Changes appear on everyone’s screen without refreshing. Owners, due dates, comments with @mentions, an activity feed. The Lead approves AI changes.

A shared lab notebook

Protocols, observations, decisions and reading notes, linked to tasks and searchable. The AI reads them when it suggests next steps.

[icon: Bell]

Nothing slips

Task owners are notified when their task is assigned, moved, commented on, or gets a result or note. Roles are enforced by the database.

PERSONA TESTING

Three researcher personas, three changes.

Persona	Likely pushback	What I changed
Lab lead environmental engineering	“Why is there a commercial case on an academic project?”	Unit economics now appears only when the goal is Product
PhD student materials science	“The plan took a minute; I almost refreshed and lost it.”	AI runs as background jobs with live progress; survives tab switches and reloads
R&D manager medical-device startup	“I can’t put unpublished ideas into a free AI tier.”	Planned: a paid AI tier whose providers don’t train on customer data

Strongest signal: **trustworthy AI**. Real papers, checkable assumptions, and directions built on the team’s own work.

Simulated personas, not real users. Real sessions with 3–5 researchers come next; their quotes will replace this slide.

THE WHOLE JOURNEY: AARRR

An answer for every stage.

A

Acquisition

Supervisors invite their students; department research groups

A

Activation

A verdict with real papers in 2 minutes; activated once a first result is logged within 7 days

R

Retention

Log results weekly; the dashboard prepares every supervisor meeting

R

Referral

Sharing a project with lab mates is part of the workflow

R

Revenue

Free for students; per-seat plans for labs and R&D teams

GO-TO-MARKET

GTM: 10 users, then 1,000, then 100k.

WHO ARE MY FIRST 10 USERS?

[6] PhD students in Prof. Manibrata Sen's Theoretical Physics lab, their [2] supervisors, and 2 R&D engineers

WHERE WILL I SHARE IT?

[Department PhD WhatsApp group], [research scholars' association], [university incubator], a slot at [lab meeting or seminar]

HOW DO I GET TO 1,000, THEN 100K?

To 1,000

Each project pulls in a supervisor and lab mates; student ambassadors in [5] departments

→

To 100k

Institution-wide licences through research offices; a free idea check as the way in

How it makes money: free for students; labs and R&D teams pay ₹399 per seat per month.

WHAT'S NEXT

Log results by voice

A WhatsApp voice note from the lab bench becomes a logged result.

Grant-ready reports

Export the dashboard and evidence for funders and progress reviews.

A "My week" view

Your tasks due this week, and partial results you can log before a task is done.

From idea to validated result, with your whole team on the same page.

Try it: research-os-beige-five.vercel.app · Code: github.com/worldofnisho-471/ResearchOS · saurrav.dasgupta@outlook.com